

Please provide a full list of commands with explanations for Git and GitHub?

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Here is a comprehensive guide to Git and GitHub commands, categorized for clarity and ease of use. This list covers essential commands, their explanations, and examples, drawing from the search results.

🧠 Git and GitHub Commands: Full List with Explanations

1. ⚙️ Configuration Commands

Command	Description	Example
<code>git config --global user.name "Your Name"</code>	Sets your username for Git commits	<code>git config</code>
<code>git config --global user.email "your@email.com"</code>	Sets your email for Git commits	<code>git config</code>
<code>git config --list</code>	Lists all Git configuration settings	<code>git config</code>

2. 📁 Repository Initialization and Cloning

Command	Description	Example
<code>git init</code>	Initializes a new Git repository	<code>git init my-proj</code>
<code>git clone <repo-url></code>	Creates a copy of an existing repository	<code>git clone https:</code>
<code>git clone <repo-url> <directory></code>	Clones a repository into a specific directory	<code>git clone https:</code>

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3. 🔄 Staging and Committing Changes

Command	Description	Example
<code>git status</code>	Shows the status of changes in the working directory	<code>git status</code> 2 8
<code>git add <file></code>	Stages a specific file	<code>git add index.html</code>
<code>git add .</code> Or <code>git add -A</code>	Stages all new and modified files	<code>git add .</code> 2 7
<code>git commit -m "message"</code>	Commits staged changes with a message	<code>git commit -m "Add index.html"</code>
<code>git commit --amend</code>	Amends the most recent commit	<code>git commit --amend</code>

4. 🌿 Branching and Merging

Command	Description	Example
<code>git branch</code>	Lists all local branches	<code>git branch</code> 2 7
<code>git branch <branch-name></code>	Creates a new branch	<code>git branch new-feature</code>
<code>git checkout <branch-name></code>	Switches to a branch	<code>git checkout main</code> 1
<code>git checkout -b <branch-name></code>	Creates and switches to a new branch	<code>git checkout -b new-feature</code>
<code>git merge <branch-name></code>	Merges a branch into the current branch	<code>git merge new-feature</code>
<code>git branch -d <branch-name></code>	Deletes a branch	<code>git branch -d old-feature</code>

5. 📁 Remote Repository Interaction

Command	Description	Example
<code>git remote add origin <url></code>	Adds a remote repository	<code>git remote add origin https://github.com/username/repository.git</code> 7 9
<code>git push -u origin <branch></code>	Pushes changes to a remote repository	<code>git push -u origin main</code>
<code>git push</code>	Pushes changes to the remembered remote branch	<code>git push</code> 2 6
<code>git pull</code>	Fetches and merges changes from the remote	<code>git pull origin main</code>
<code>git fetch</code>	Downloads changes but doesn't merge	<code>git fetch origin</code>

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6. Viewing History and Differences

Command	Description	Example
<code>git log</code>	Shows commit history	<code>git log</code> 1 8
<code>git log --oneline</code>	Shows concise commit history	<code>git log --oneline</code> 7
<code>git diff</code>	Shows unstaged changes	<code>git diff</code> 8 10
<code>git diff --staged</code>	Shows staged changes	<code>git diff --staged</code> 10
<code>git show <commit></code>	Shows changes in a specific commit	<code>git show abc123</code> 2

7. Undoing Changes

Command	Description	Example
<code>git restore <file></code>	Discards changes in the working directory	<code>git restore file.txt</code> 8
<code>git reset <file></code>	Unstages a file	<code>git reset file.txt</code> 1 8
<code>git revert <commit></code>	Reverts a specific commit	<code>git revert abc123</code> 1 7

8. Stashing Changes

Command	Description	Example
<code>git stash</code>	Temporarily saves changes	<code>git stash</code> 4 7
<code>git stash pop</code>	Restores stashed changes	<code>git stash pop</code> 4
<code>git stash list</code>	Lists all stashes	<code>git stash list</code> 4

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9. 🏷️ Tagging

Command	Description	Example
<code>git tag <tag-name></code>	Creates a new tag	<code>git tag v1.0</code> 1 4
<code>git push --tags</code>	Pushes tags to the remote repository	<code>git push --tags</code> 4

10. 🔍 Advanced Commands

Command	Description	Example
<code>git rebase <branch></code>	Reapplies commits on top of another branch	<code>git rebase main</code>
<code>git cherry-pick <commit></code>	Applies a specific commit to the current branch	<code>git cherry-pick</code>
<code>git bisect</code>	Uses binary search to find a bug-introducing commit	<code>git bisect start</code>

11. 📖 GitHub-Specific Commands

Command	Description	Example
<code>git pull --rebase</code>	Rebases local changes on top of remote changes	<code>git pull --rebase origin m</code>
<code>git push --force</code>	Force pushes changes (use with caution)	<code>git push --force origin ma</code>

12. 💡 Best Practices and Tips

- **Commit Messages:** Write clear, concise commit messages. The first line should be a short summary (50 characters or less), followed by a blank line and a detailed description if needed 10 .
- **Branch Naming:** Use descriptive branch names, such as `feature/add-new-button` or `fix/issue-123` 2 .
- **Pull Before Push:** Always pull the latest changes from the remote before pushing your own to avoid conflicts 2 9 .
- **Use .gitignore:** Add a `.gitignore` file to exclude unnecessary files (e.g., log files, dependencies) from being tracked 9 .

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13. 🚀 Common Workflows

- Feature Branch Workflow:

1. Create a new branch: `git checkout -b new-feature`
2. Make changes and commit: `git commit -m "Add new feature"`
3. Push the branch: `git push -u origin new-feature`
4. Open a pull request on GitHub ¹ ² .

- Forking Workflow:

1. Fork a repository on GitHub.
2. Clone your fork: `git clone https://github.com/your-username/repo.git`
3. Add the original repository as a remote:
`git remote add upstream https://github.com/original/repo.git`
4. Fetch updates: `git fetch upstream`
5. Merge updates: `git merge upstream/main` ⁹ .

14. 🆘 Getting Help

- Use `git help <command>` to view the manual for a specific command ¹⁰ .
- Use `git <command> --help` for detailed information about a command.

15. 📚 Additional Resources

- [Official Git Documentation](#)
- [GitHub Documentation](#) ⁹
- [Git Tutorials on W3Schools](#) ³

This list covers the most essential Git and GitHub commands. For a more comprehensive reference, you can explore the provided sources or check out cheat sheets from [GitKraken](#) ⁴ and [LoginRadius](#) ⁷ .

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